

SEMESTER END EXAMINATIONS – AUGUST 2024

Program	: B.E. – Information Science and Engineering	Semester	: IV
Course Name	: Advanced Java	Max. Marks	: 100
Course Code	: IS45	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What are the two main objectives of Generics? Give example for each objective. CO1 (08)
 - Write a java code to check the occurrence of a substring in a given string. CO1 (06)
 - Explain any three character handling functions in string class. CO1 (06)
- Explain the following built in methods with respect to string buffer class: CO1 (10)
 - capacity()
 - reverse()
 - replace()
 - insert()
 - append()
 - Write a java program to compare the difference between String using == and equals() method in Java. CO1 (06)
 - Explain the difference between String Class and String Buffer class. CO1 (04)

UNIT - II

- Infer the role of the Object class in Java and how it serves as the root of the class hierarchy. CO2 (06)
 - State the significance of comparing objects in Java, including methods such as equals(), hashCode(), and compareTo(). CO2 (07)
 - Define functional interfaces in Java and explain their role in enabling lambda expressions and functional programming paradigms. CO2 (07)
- Define an iterator in Java, and how does it facilitate traversal of collections? Provide an example. CO2 (06)
 - Summarize the advantages of using method references in Java compared to lambda expressions, highlighting scenarios where each approach is beneficial. CO2 (07)
 - Examine the purpose of wrapper classes in Java and how autoboxing simplifies the conversion between primitive types and their corresponding object types. CO2 (07)

UNIT - III

5. a) Explain two ways of creating threads in Java, with an example. CO3 (10)
b) Consider the following Java code: CO3 (10)
`MyRunnable mr = new MyRunnable(); //MyRunnable implements Runnable`
`Thread t1 = new Thread ();`
`Thread t2 = new Thread(mr);`
Analyze the following method calls and describe their response:
i. `t1.start()`
ii. `t2.start()`
iii. `mr.start()`
iv. `mr.run()`
6. a) List five methods of Collection Interface. Write java code for CO3 (10)
demonstrating their usage.
b) Write Java program to sort the given ArrayList in Descending order. CO3 (05)
c) Write Java code to insert integer objects into TreeSet where the sorting CO3 (05)
order is descending.

UNIT- IV

7. a) List and explain the life cycle of servlet and its associated methods. CO4 (10)
b) Write servlet code to read records from Products (pno, pname, pprice) CO4 (10)
table in mysql database and display them on a web page. Use web.xml
descriptor to identify servlet.
8. a) Illustrate web application directory structure and list the contents of each CO4 (08)
folder.
b) List any three annotations related to servlets in Java and demonstrate CO4 (07)
their usage with code snippets.
c) Differentiate between Get and Post methods. CO4 (05)

UNIT- V

9. a) What is JDBC? Explain the different JDBC driver types. CO5 (10)
b) Illustrate the life cycle of a JSP page and give example code to CO5 (06)
demonstrate each component.
c) List the kinds of exceptions occurring in JDBC. CO5 (04)
10. a) With a code snippet, explain the different steps involved in JDBC. CO5 (12)
b) Explain different JSP tags with a program to demonstrate all tags. CO5 (08)
